QQ-A-250/2E
8 July 1983
SUPERSEDING
QQ-A-250/2D
November 23, 1970

FEDERAL SPECIFICATION SHEET

ALUMINUM ALLOY 3003, PLATE AND SHEET

This specification was approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal agencies.

The complete requirements for procuring the aluminum alloy 3003 plate and sheet described herein shall consist of this document and the latest issue of the general specification QQ-A-250 (see 2.1).

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the specific requirements for aluminum alloy 3003 plate and sheet.

1.2 Classification.

1.2.1 Tempers. The plate and sheet shall be of the following tempers: O, H12, H14, H16, H18, H22, H24, H26, H28, H112 or F, as specified (see 6.2). The definition of these tempers shall be as specified in the general specification QQ-A-250.

1.2.1.1 Temper substitutions. Material in either one of a pair of tempers (H12 and H22, H14 and H24, H16 and H26, or H18 and H28) may be supplied at the option of the producer or supplier, unless the supplying of one temper of the pair is expressly forbidden in the procurement documents (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Government publications. The issues of the following documents, in effect on date of invitation for bids or solicitation for offers, form a part of this specification to the extent specified herein.

Federal Specifications

QQ-A-250 - Aluminum and Aluminum Alloy, Plate and Sheet; General Specification for

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions, as outlined under

FSC 9535

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General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY: Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

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3. REQUIREMENTS

3.1 Chemical composition. The chemical composition shall conform to the requirements specified in table I.

TABLE I. Chemical composition 1/

Element	Percent	
	Minimum	Maximum
Silicon	--	0.6
Iron	--	0.7
Copper	0.05	0.20
Manganese	1.0	1.5
Zinc	--	0.10
Others, each	--	0.05
Others, total 2/	--	0.15
Aluminum	Remainder	

1/ Analysis shall routinely be made only for the elements specifically mentioned in table I. If, however, the presence of other elements is indicated or suspected in amounts greater than the specified limits, further analysis shall be made to determine that these elements are not present in excess of specified limits

2/ The sum of those "Others" metallic elements 0.010 percent or more each, expressed to the second decimal before determining the sum.

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3.2 Mechanical properties.

3.2.1 Mechanical properties of material as supplied. The mechanical properties parallel to the direction of the final rolling shall conform to the requirements specified in table II for the temper specified.

TABLE II. Mechanical properties

Temper	Thickness, inches	Ultimate tensile strength, ksi		Yield strength at 0.2 percent offset,	Elongation in 2 inches or 4 times D 2/, 3/
		Minimum	Maximum	Minimum, ksi 1/	Minimum, percent
O	0.006-0.007	14.0	19.0	5.0	14
	0.008-0.012	14.0	19.0	5.0	18
	0.013-0.031	14.0	19.0	5.0	20
	0.032-0.050	14.0	19.0	5.0	23
	0.051-0.249	14.0	19.0	5.0	25
	0.250-3.000	14.0	19.0	5.0	23
H12	0.017-0.019	17.0	23.0	12.0	3
	0.020-0.031	17.0	23.0	12.0	4
	0.032-0.050	17.0	23.0	12.0	5
	0.051-0.113	17.0	23.0	12.0	6
	0.114-0.161	17.0	23.0	12.0	7
	0.162-0.249	17.0	23.0	12.0	8
	0.250-0.499	17.0	23.0	12.0	9
	0.500-2.000	17.0	23.0	12.0	10
H14	0.009-0.012	20.0	26.0	17.0	1
	0.013-0.019	20.0	26.0	17.0	2
	0.020-0.031	20.0	26.0	17.0	3
	0.032-0.050	20.0	26.0	17.0	4
	0.051-0.113	20.0	26.0	17.0	5
	0.114-0.161	20.0	26.0	17.0	6
	0.162-0.249	20.0	26.0	17.0	7
	0.250-0.499	20.0	26.0	17.0	8
	0.500-1.000	20.0	26.0	17.0	10
H16	0.006-0.019	24.0	30.0	21.0	1
	0.020-0.031	24.0	30.0	21.0	2
	0.032-0.050	24.0	30.0	21.0	3
	0.051-0.162	24.0	30.0	21.0	4

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TABLE II. Mechanical properties (CON)

Temper	Thickness, inches	Ultimate tensile strength, ksi		Yield strength at 0.2 percent offset,	Elongation in 2 inches or 4 times D 2/, 3/
		Minimum	Maximum	Minimum, ksi 1/	Minimum, percent
H18	0.006-0.019	27.0	--	24.0	1
	0.020-0.031	27.0	--	24.0	2
	0.032-0.050	27.0	--	24.0	3
	0.051-0.128	27.0	--	24.0	4
H112	0.250-0.499	17.0	--	10.0	8
	0.500-2.000	15.0	--	6.0	12
	2.001-3.000	14.5	--	6.0	18
H22	0.017-0.019	17.0	--	--	3
	0.020-0.031	17.0	--	--	4
	0.032-0.050	17.0	--	--	5
	0.051-0.113	17.0	--	--	6
	0.114-0.161	17.0	--	--	7
	0.162-0.249	17.0	--	--	8
	0.250-0.499	17.0	--	--	9
	0.500-2.000	17.0	--	--	10
H24	0.009-0.012	20.0	--	--	1
	0.013-0.019	20.0	--	--	2
	0.020-0.031	20.0	--	--	3
	0.032-0.050	20.0	--	--	4
	0.051-0.113	20.0	--	--	5
	0.114-0.161	20.0	--	--	6
	0.162-0.249	20.0	--	--	7
	0.250-0.499	20.0	--	--	8
	0.500-1.000	20.0	--	--	10
H26	0.006-0.019	24.0	--	--	1
	0.020-0.031	24.0	--	--	2
	0.032-0.050	24.0	--	--	3
	0.051-0.162	24.0	--	--	4
H28	0.006-0.019	27.0	--	--	1
	0.020-0.031	27.0	--	--	2
	0.032-0.050	27.0	--	--	3
	0.051-0.128	27.0	--	--	4

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TABLE II. Mechanical properties (CON)

Temper	Thickness, inches	Ultimate tensile strength, ksi		Yield strength at 0.2 percent offset,	Elongation in 2 inches or 4 times D 2/, 3/
		Minimum	Maximum	Minimum, ksi 1/	Minimum, percent
F	All	4/	4/	4/	4/

1/ 1.000 ksi equals 1,000 psi
2/ Not required for material 0.500 inch or less in width
3/ "D" is specimen diameter
4/ No requirements

3.2.2 Bend properties. Bend-test specimens taken from material shall be capable of withstanding, without cracking, the bend test specified in the general specification QQ-A-250. The values for bend factor "N" are listed in table III.

TABLE III. Bend test factor "N"

Thickness of material, inches	Temper			
	O	H12 or H22	H14 or H24	H16 or H26
0.006-0.009	0	--	0	4
0.010-0.016	0	--	0	4
0.017-0.049	0	0	0	4
0.050-0.113	0	0	0	6
0.114-0.162	0	0	2	6
0.163-0.249	0	0	2	--

4. QUALITY ASSURANCE PROVISIONS (see the general specification QQ-A-250.

4.1 Sampling. Sampling for conformance to the requirements of section 3 shall be as specified in the general specification QQ-A-250.

5. PREPARATION FOR DELIVERY (see the general specification QQ-A-250)

6. NOTES

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6.1 Intended use. Alloy 3003 is a general purpose aluminum alloy for moderate strength applications. It has good workability characteristics, is readily welded, and has good corrosion resistance.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number and date of this specification
(b) Form, dimensions, quantity and temper required (see 1.2.1)
(c) If substitution for temper ordered is forbidden (see 1.2.1.1)
(d) Requirements for sizes not specifically covered by this detail specification (see the general specification QQ-A-250)
(e) Special end use requirements
(f) Selection of applicable levels of preservation/packaging and packing required, if other than level C (see the general specification QQ-A-250)

MILITARY INTEREST:

Custodians
Army-MR
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Air Force-99

User Activities
Army-CR, ME
Navy-MC

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA-FSS, PBO
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PREPARING ACTIVITY:

NAVY-AS

DOD project 9535-0341

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
QQ-A-250/2E	Aluminum Alloy 3003, Plate and Sheet

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

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c. Reason/Rationale for Recommendation:

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7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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